

Technology News

Robot with emotions

Researchers at the University of Hertfordshire have designed a robot that not only shows human emotions but also develops them on its own. Nao imitates the social and emotional skills of a one-year-old child; the humanoid machine looks down when it feels sad and raises its arms angling for a hug to express happiness.

Nao needs to 'know' a person, study his body language and facial expressions to detect his emotions. The more the interaction with someone, the more adept the robot at reading a person's mood. Nao makes use of video cameras to detect how close a person comes and sensors to determine how people are interacting with him and act accordingly.

The robot is part of Felix Grow-

ing—a European project that aims at creating companions who can live with people and support them in daily activities.

Mind-reading computer

Imagine searching over the Internet just by thinking? Well, if Intel's scientists are to be believed, a new technology will directly interpret words as they are thought, allowing people to dictate letters and search the Internet simply by thinking.

Intel scientists are creating detailed maps of the activity in the brain for individual words, which can then be matched against the brain activity of someone using the computer, allowing the machine to determine the words they are thinking.

Dean Pomerleau, a senior researcher at Intel Laboratories, said: "The computer uses a form of 20 questions to narrow down what the word is. So a noun with a physical property such as spade, which you dig with, produces activity in the motor cortex of the brain, as this is the area that controls physical movements."

Now touch and feel the virtual world

It may sound like a science fiction, but researchers at the Computer Vision Lab at ETH Zurich, Switzerland, have developed a new technology that allows you to touch the virtual world. The technology helps researchers produce virtual copies of real objects, which can not only be touched but also sent via the Internet, eventually allowing you to meet the desired person online.

The researchers devel-

oped a method for combining visual and haptic impressions with one another. Whilst a 3-D scanner records an image of the object, which in one experiment was a soft toy frog, a user simultaneously senses the object using a haptic device. The sensor arm, which can be moved in any direction and is equipped with force, acceleration and slip sensors, collects information about shape and solidity. With the aid of an algorithm, a virtual copy is created on the computer from the measurements—even while the toy frog is still being scanned and probed.

The virtual copy, sent over the Internet, can be seen using data goggles with a monitor and a sensor rod equipped with small motors.

Although scientists have already developed a technology that makes it possible to touch a virtual object, this one allows you to both touch and grasp the object as special sensor gloves let you first 'scan in' a real object and then touch a virtual object.

Turn any surface into touchscreen

Now you can convert your devices like mobile phone and TV into touchscreen ones with a miracle technology from an interactive hardware company



Nao robot



Displax Skin technology